Hello Scarlet,

The specification requirements regarding similar shapes in GCSE (9-1) Mathematics are as follows:

R12 compare lengths, areas and volumes using ratio notation; make links to similarity (including trigonometric ratios) and scale factors

G19 apply the concepts of congruence and similarity, including the relationships between lengths, areas and volumes in similar figures

Therefore, although students may be taught how to prove the relationship between the linear and area scale factors in order to develop their conceptual understanding, questions tend to focus on application rather than proof.

The GCSE question that is most similar to the one you describe:

I have described a Tenth standard Indian maths question (equivalent to British year 11)

The question is "prove that the ratio of the areas of two similar triangles is equal to the ratio of the squared lengths of two corresponding sides of the triangles"

is Q9, Paper 2H, June 2019:

The circumference of circle **B** is 90% of the circumference of circle **A**.

(a) Find the ratio of the area of circle **A** to the area of circle **B**.

I hope that this information is helpful. Please do not hesitate to get back in touch should you have any further questions.
Best regards
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